SUPPORTING ADULT GUIDANCE

Area of rectangles and triangles

Perimeter and Area · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, Cut and Turn, learners first drag a rectangle on a 1 m grid and see its squares in rows, so counting becomes length × width. They then drag a triangle's top corner to any grid point and its base along the line; the rectangle with the same base and height is shown dashed, with the perpendicular height in red. Cut and turn splits the triangle along its height and turns the two pieces half a turn into the empty corners, showing the triangle is exactly half. For obtuse triangles, Show why explains the area as a big right-angled triangle minus a small one. Six questions and a stretch follow; learners then find areas quickly, judge statements and make triangles to order.

Driving Question: *Why is a triangle half a rectangle?*

The mathematics, in plain terms

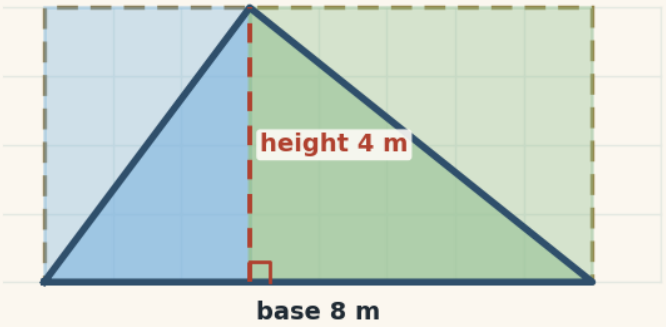
Area of a rectangle = length × width; area of a triangle = ½ × base × perpendicular height. The cut-and-turn argument uses the same half-turn idea as Angles Lesson 3. Corners sit on whole-metre grid points, so every area shown is exact, a whole number or a half. For obtuse triangles the height lies outside, and the area is the difference of two right-angled triangles. Parallelograms and trapezia follow in Lesson 3.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Cut and turn

Find the area of a triangle with base 8 m and height 4 m.

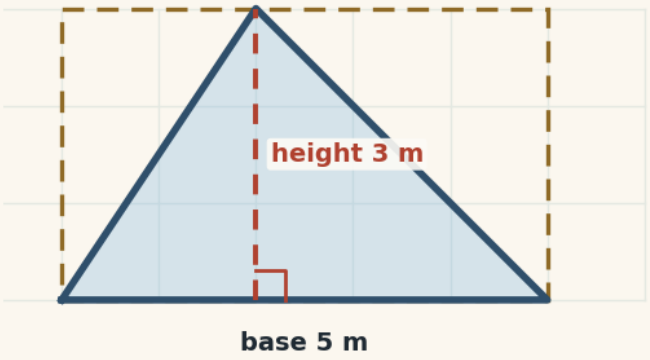


1. The rectangle round it: 8 × 4	32 m^2
2. Cut along the height; turn each piece into the empty corner beside it.	the pieces fill the rest exactly
3. So the triangle is half the rectangle.	½ × 32 = 16 m^2

Area = ½ × 8 × 4 = 16 m^2.

Example 2 — A half in the answer

Find the area of a triangle with base 5 m and height 3 m.



1. Base × height — $5 \times 3 = 15$
2. Halve it. — $15 \div 2 = 7½$
3. Area — $7½$ m^2 (or 7.5 m^2)

Area = 7½ m^2.

Rows, then halves, then the reason

Learners see rows of squares become length × width before triangles appear, then watch the triangle fill half its rectangle for many shapes before the rule is stated. The stretch makes the half-turn argument explicit, and the obtuse case guards against using a slanted side.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Cut and Turn: rectangles, then triangles, with six questions. Stretch: why every triangle is half its rectangle.	~16 min
Movement	Quick areas.	~7 min
Reflection	True or not? Which length is the height?	~7 min
Creativity	Make a triangle with a given area.	~10 min
Rest	Half, every time.	~2 min
Nutrition	Intellectual: building on what you know.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using a slanted side as the height.
Ask: “Is that line at right angles to the base?”

Forgetting to halve for a triangle.
Ask: “What would the rectangle round it be? How much of it does the triangle fill?”

Giving the perimeter instead of the area of a rectangle.
Ask: “Are you going round the edge, or counting squares inside?”

Rounding a half (7½ written as 8).
Ask: “What is half of 15 exactly?”

Thinking an obtuse triangle has no height.
Ask: “Can you extend the base line until the height meets it?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: perimeter or area (Lesson 1), and multiplying. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner gives the area of a triangle with base 6 m and slanted side 5 m as 15 m^2. The aim is a question that brings back the perpendicular height.

- “Which line did you use for the height?”
- “Is it at right angles to the base?”

- “Where would the rectangle round the triangle be?”
- “How tall is that rectangle?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why ½ × base × height and not base × height ÷ 2?
They are the same. Learners may use either; the picture (half a rectangle) matters more than the order.

Should answers with halves be written 7½ or 7.5?
Either. The lesson accepts 7½, 7.5 and 15/2.

Is the stretch required?
No. The secure core is length × width for rectangles and ½ × base × height for triangles, with the right height.

What comes next?
Lesson 3: area of parallelograms and trapezia.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.